Dr. Ch. Sobhan Babu, Ph.D.

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Areas of Interest

Big Data Analytics, Financial Fraud Analytics, Non-compliance Detection, Applied Graph Theory.

Administrative Work

- Instrumental in setting up the executive M.Tech programme and acted as incharge for initial two years.
- Developed, maintaining, and enhancing AIMS, a comprehensive system that manages the entire student life cycle — from application and admission to convocation.

Employment History

- 2016 – Till date: Associate Professor, Department of Computer Science & Engineering, IIT Hyderabad.
- 2010 – 2015: Assistant Professor, Department of Computer Science & Engineering, IIT Hyderabad.
- 2008 – 2009: Post Doctoral Fellow, Department of Computer Science & Engineering, IIT Bombay.
- 2007 – 2007: Senior Software Engineer, Softjin Technologies, Bangalore, India.
- 2001 – 2001: Senior Software Engineer, Tessera Systems, Hyderabad, India.

Education

- 2002 – 2006: Ph.D., IIT Bombay.
  Thesis title: Degree Conditions for Subgraphs.
- 1999 – 2001: M.Tech., Computer Science, IIT Bombay.
- 1996 – 1999: B.E., Computer Science, University of Madras.
- 1991 – 1995: Diploma in Computer Engineering, SBTET, Hyderabad.

Other National Level Services

- IT Adviser to Insurance Regulatory and Development Authority of India.
- Providing Support to Ayushman Bharat Digital Mission (ABDM), Ministry of Health, Government of India.
- Assisting Bank of India (BOI) to host natiuonal level FinTech & Cybersecurity Hackathon Series. This initiative is powered by the Department of Financial Services (DFS) (Ministry of Finance)/Indian Bank Association (IBA).

Research Publications

Journal Articles

1. Damalla, R., Gayathri, C., Datla, R., & Ch. Sobhan Babu. (2025). SuperCLIP: Semantic Attribute-Guided Transformer With Super-Resolution and CLIP for Zero-Shot Remote Sensing Scene Classification. IEEE Geoscience and Remote Sensing Letters, 22, 1–5.

2. Swetha, G., Datla, R., Chintapalli, S. B., & Krishna Mohan, C. (2025). CPL-PL: Contrapositive Learning-Based Pseudo-Labeling for Semi-Supervised Scene Classification in Remote Sensing Images. IEEE Geoscience and Remote Sensing Letters, 22, 1–5.

3. Vishnu, C., Abhinav, V., Roy, D., Mohan, C. K., & Babu, C. S. (2023). Improving multi-agent trajectory prediction using traffic states on interactive driving scenarios. IEEE Robotics and Automation Letters, 8(5), 2708–2715.

4. Chalavadi, V., Prudviraj, J., Datla, R., Babu, C. S., & C, K. M. (2022). Msodanet: A network for multi-scale object detection in aerial images using hierarchical dilated convolutions. Pattern Recognition, 126, 108548.

5. Bangaru, M. L. Y., Medabalimi, R. K., Babu, S., & Raghavendra, N. K. (2021). REMP software to introduce a screening restriction site in site-directed mutagenesis primer. SoftwareX, 16, 100881.

6. Vishnu, C., Datla, R., Roy, D., Babu, S., & Mohan, C. K. (2021). Human fall detection in surveillance videos using fall motion vector modeling. IEEE Sensors Journal, 21(15), 17162–17170.

7. Adireddy, S. T., Potluri, S., Babu, C. S., Kamakoti, V., & Singh, S. G. (2017). Optimal don't care filling for minimizing peak toggles during at-speed stuck-at testing. ACM Transactions on Design Automation of Electronic Systems, 23(1), 5:1–5:26.

8. Potluri, S., Trinadh, S., Babu, C. S., Kamakoti, V., & Chandrachoodan, N. (2015). DFT assisted techniques for peak launch-to-capture power reduction during launch-on-shift at-speed testing. ACM Transactions on Design Automation of Electronic Systems, 21(1), 14:1–14:25.

9. Trinadh, S., Potluri, S., Balachandran, S., Babu, C. S., & Kamakoti, V. (2014). Xstat: Statistical X-filling algorithm for peak capture power reduction in scan tests. Journal of Low Power Electronics, 10(1), 107–115.

10. Trinadh, S., Potluri, S., Babu, C. S., & Kamakoti, V. (2013). An efficient heuristic for peak capture power minimization during scan-based test. Journal of Low Power Electronics, 9(2), 264–274.

11. Guravannavar, R., Sudarshan, S., Diwan, A. A., & Babu, C. S. (2012). Which sort orders are interesting? VLDB Journal, 21(1), 145–165.

12. Babu, C. S., & Diwan, A. A. (2009). Disjoint cycles with chords in graphs. Journal of Graph Theory, 60(2), 87–98.

13. Babu, C. S., & Diwan, A. A. (2008a). Degree condition for subdivisions of unicyclic graphs. Graphs and Combinatorics, 24(6), 495–509.

14. Babu, C. S., & Diwan, A. A. (2008b). Subdivisions of graphs: A generalization of paths and cycles. Discrete Mathematics, 308(19), 4479–4486.

15. Babu, C. S., & Diwan, A. A. (2005a). Degree conditions for forests in graphs. Discrete Mathematics, 301(2–3), 228–231.

16. Babu, C. S., & Diwan, A. A. (2005b). Oriented forests in directed graphs. Electronic Notes in Discrete Mathematics, 22, 141–145.

Conference Proceedings

1. Peketi, D., Krishna Mohan, C., Yenduri, S., & Sobhan Babu, C. (2025). FSF3A: Federated Spatial Feature Alignment and Adaptive Aggregation for Heterogeneous Brain Tumor Segmentation. Accepted in British Machine Vision Conference (BMVC), 2025.

2. Subbareddy, B., Pamisetty, G., Kothuri, M., Verma, P., & Ch. Sobhan Babu. (2024). Adaptive Neighborhood Sampling and KAN-Based Edge Features for Enhanced Fraud Detection. IEEE Big Data 2024: 3375–3381.

3. Pamisetty, G., Batreddy, S., Verma, P., & Ch. Sobhan Babu. (2024). EHRGPT: Leveraging language models for generating synthetic health records. IEEE Big Data 2024: 4579–4587.

4. Mehta, P., Bhargava, S., Kumar, K. S., Kumar, M. R., & Babu, C. S. (2022). Representation learning on graphs to identify circular trading in goods and services tax. IEEE International Conference on Big Data (Big Data 2022), Osaka, Japan.

5. Mehta, P., Kumar, S., Kumar, R., & Babu, C. S. (2022). Enhancement to training of bidirectional GAN: An approach to demystify tax fraud. IEEE International Conference on Big Data (Big Data 2022), Osaka, Japan.

6. Bhargava, S., Kumar, M. R., Mehta, P., Mathews, J., Kumar, K. S., & Babu, C. S. (2021). A novel example-dependent cost-sensitive stacking classifier to identify tax return defaulters. 24th International Conference on Business Information Systems (BIS 2021), Hannover, Germany.

7. Mathews, J., Mehta, P., Suryamukhi, K., & Babu, C. S. (2021). Link prediction techniques to handle tax evasion. CODS-COMAD 2021: 8th ACM IKDD CODS and 26th COMAD, Bangalore, India.

8. Mehta, P., Kumar, K. S., Kumar, R., & Babu, C. S. (2021). Demystifying tax evasion using variational graph autoencoders. EGOVIS 2021, Springer.

9. Mehta, P., Mathews, J., Bisht, D., Suryamukhi, K., Kumar, K. S., & Babu, C. S. (2020). Detecting tax evaders using TrustRank and spectral clustering. BIS 2020, Colorado Springs, USA.

10. Mehta, P., Rao, S. V. K. V., Kumar, K. S., & Babu, C. S. (2020). Deepcatch: Predicting return defaulters in taxation system using example-dependent cost-sensitive deep neural networks. IEEE BigData 2020.

11. Mehta, P., Mathews, J., Kasi Visweswara Rao, S. V., Kumar, K. S., Suryamukhi, K., & Babu, C. S. (2019). Identifying malicious dealers in goods and services tax. 2019 IEEE 4th International Conference on Big Data Analytics (ICBDA).

12. Mehta, P., Mathews, J., Kumar, K. S., Suryamukhi, K., & Babu, C. S. (2019). Curtailing the tax leakages by nabbing return defaulters in taxation system. AusDM 2019, Adelaide, Australia.

13. Mehta, P., Mathews, J., Kumar, K. S., Suryamukhi, K., Babu, C. S., & Rao, S. V. K. V. (2019). Big data analytics for nabbing fraudulent transactions in taxation system. BigData 2019, SCF 2019, San Diego, USA.

14. Mehta, P., Mathews, J., Kumar, K. S., Suryamukhi, K., Babu, C. S., Rao, S. V. K. V., Shivapujimath, V., & Bisht, D. (2019). Big data analytics for tax administration. EGOVIS 2019, Linz, Austria.

15. Priya, Mathews, J., Kumar, K. S., Babu, C. S., & Rao, S. V. K. V. (2019). A collusion set detection in value added tax using Benford's analysis. Intelligent Computing, Springer.

16. Mathews, J., Mehta, P., Babu, C. S., & Rao, S. V. K. V. (2018). Clustering collusive dealers in commercial taxation system. INTELLISYS 2018, London, UK.

17. Mathews, J., Mehta, P., Kuchibhotla, S., Bisht, D., Chintapalli, S. B., & Rao, S. V. K. V. (2018). Regression analysis towards estimating tax evasion in goods and services tax. IEEE/WIC/ACM Web Intelligence 2018, Santiago, Chile.

18. Mehta, P., Mathews, J., Suryamukhi, K., Kumar, K. S., & Babu, C. S. (2018). Predictive modeling for identifying return defaulters in goods and services tax. IEEE DSAA 2018, Turin, Italy.

19. Vishnu, C., Singh, D., Mohan, C. K., & Babu, S. (2017). Detection of motorcyclists without helmet in videos using convolutional neural network. IJCNN 2017, Anchorage, USA.

20. Trinadh, S., Babu, C. S., Singh, S. G., Potluri, S., & Kamakoti, V. (2015). DP-Fill: A dynamic programming approach to X-filling for minimizing peak test power in scan tests. DATE 2015, Grenoble, France.

21. Adsul, B., Babu, C. S., Garg, J., Mehta, R., & Sohoni, M. A. (2010a). Nash equilibria in Fisher market. SAGT 2010, Athens, Greece.

22. Adsul, B., Babu, C. S., Garg, J., Mehta, R., & Sohoni, M. A. (2010b). A simplex-like algorithm for Fisher markets. SAGT 2010, Athens, Greece.